



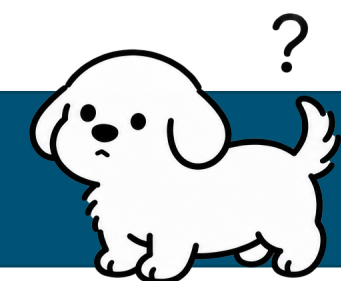
QDA-SQL: Questions Enhanced Dialogue Augmentation for Multi-Turn Text-to-SQL

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Why multi-turn needs more than SQL



A useful assistant must know when to query, clarify, or stop.

Most benchmarks test one turn at a time, but real users revise constraints, refer to earlier answers, or ask questions SQL cannot answer. Without context and intent recognition, a model may produce plausible but unsupported SQL.

ANSWERABLE

Use SQL to retrieve evidence and answer with database-grounded facts.

AMBIGUOUS

Ask a focused follow-up when the request has multiple valid readings.

UNANSWERABLE

Explain what evidence is missing; never fabricate a database result.

IMPROPER

Respond naturally when the request does not call for a SQL query.

Most Text-to-SQL data covers one answerable turn, leaving models weak at preserving context and recognizing ambiguity. QDA-SQL therefore expands SQL goals into intent-labeled multi-turn dialogues, then verifies each sample—teaching models what to query, when to clarify, and when to stop.

QDA-SQL generation: diversify, verify, refine

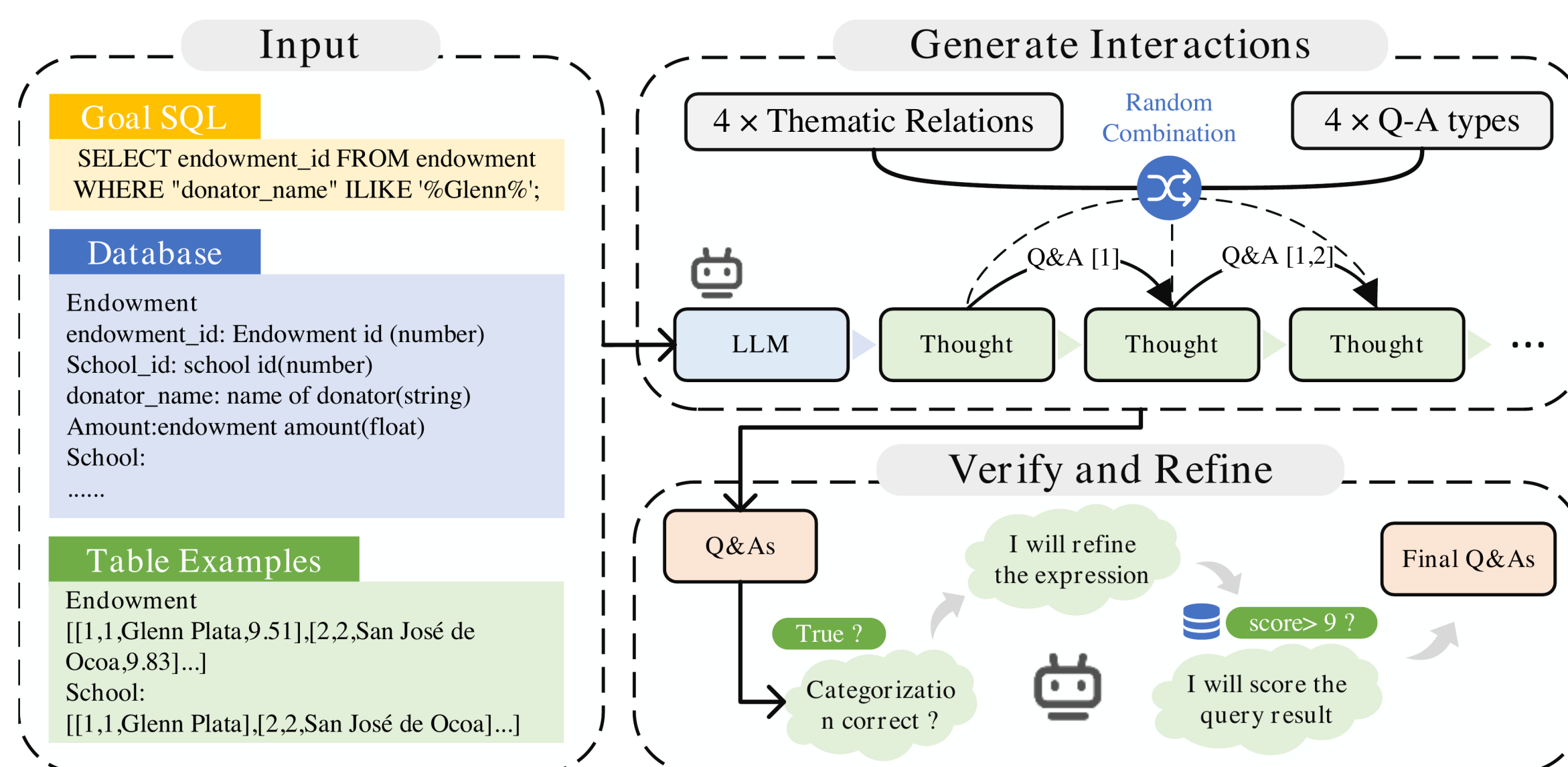
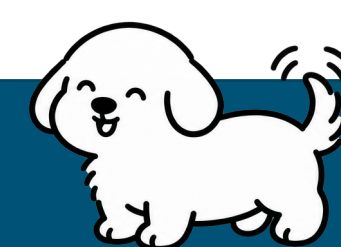
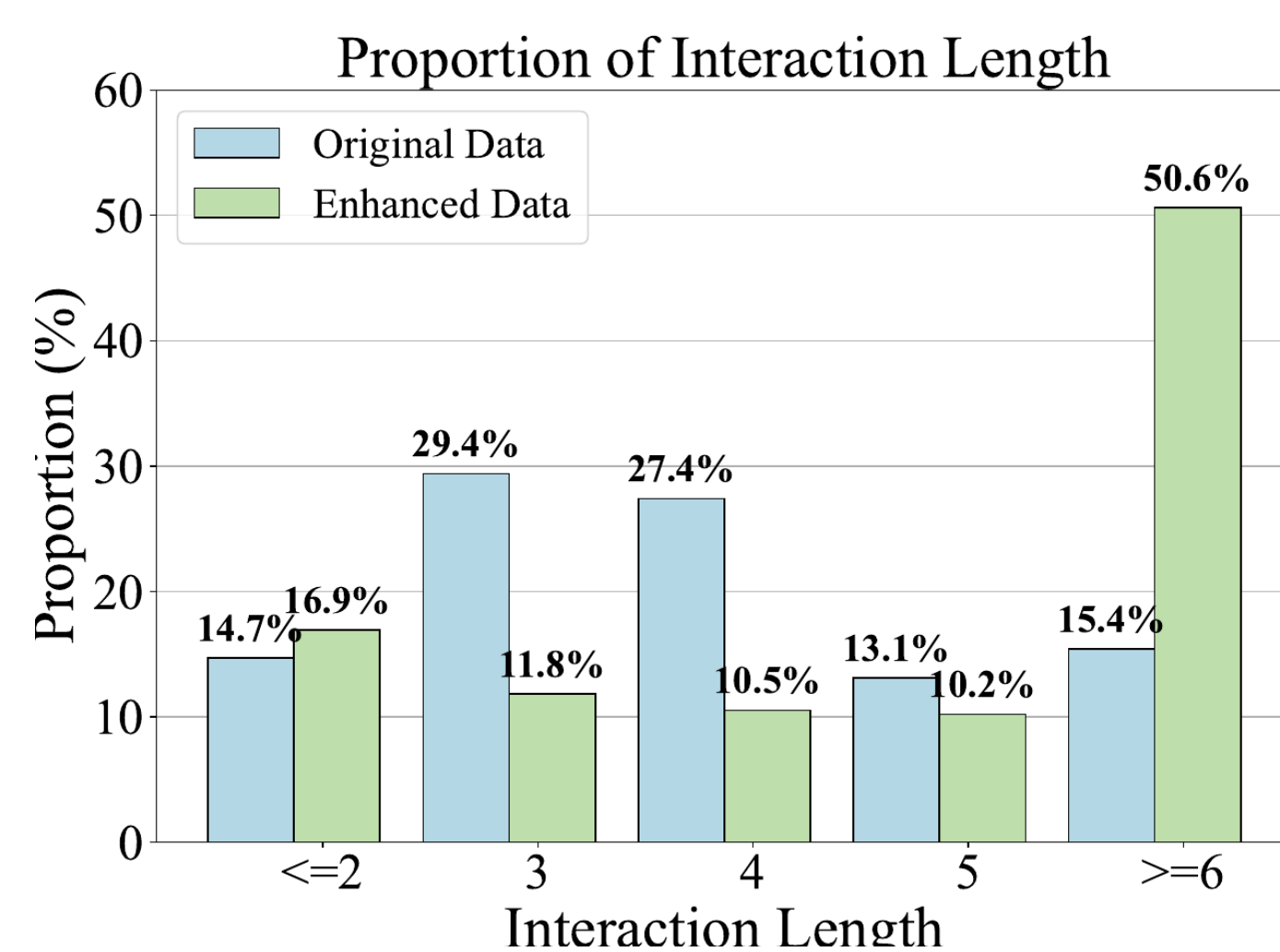


Fig. 2. Overview of QDA-SQL processes.



Start with Goal SQL, schema, and example rows. Generate follow-up questions using four thematic relations, then validate and refine the dialogue so each final sample remains executable. The database-and-paw mascot marks this data-grounded starting point.

More turns, broader coverage



10,874

generated dialogue sessions from SQL goals and schemas

65,393

question-and-response turns for multi-turn training

6.0

average turns per dialogue beyond one-shot SQL

Four thematic relations shape what the assistant asks next:

Constraint refinement

Add a filter or tighten a limit.

Topic exploration

Ask about a related subject.

Participant shift

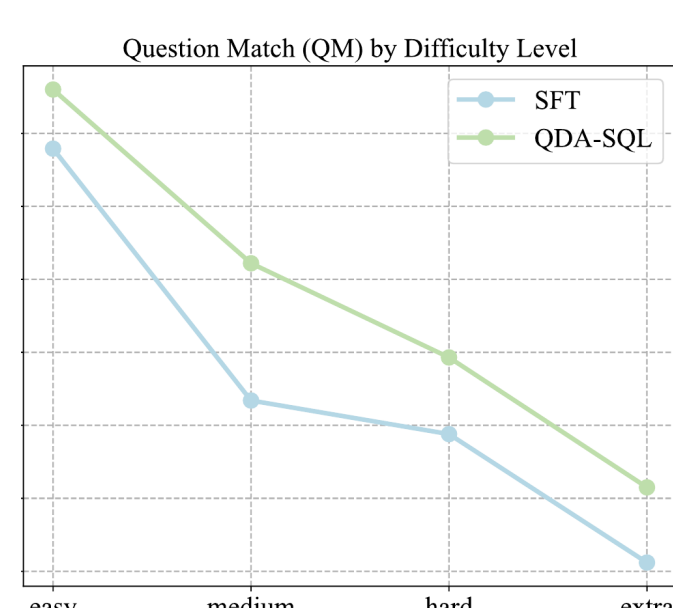
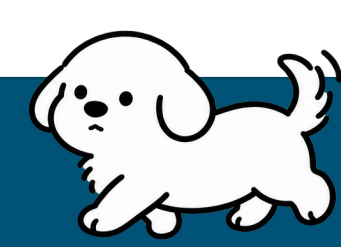
Change the person or entity.

Answer exploration

Request detail on an earlier answer.

These relations preserve context while broadening coverage. Goal SQL still relies on human annotation, and generated dialogues may miss real-world scenarios.

Harder SQL, stronger preference



QDA-SQL maintains higher Question Match from easy through extra-hard SQL.

62%

Preferred overall; one quarter of pairwise judgments were ties.

94%

47 of 50 initially misclassified samples were recovered after refinement.

StateFlow: recognize · solve · verify

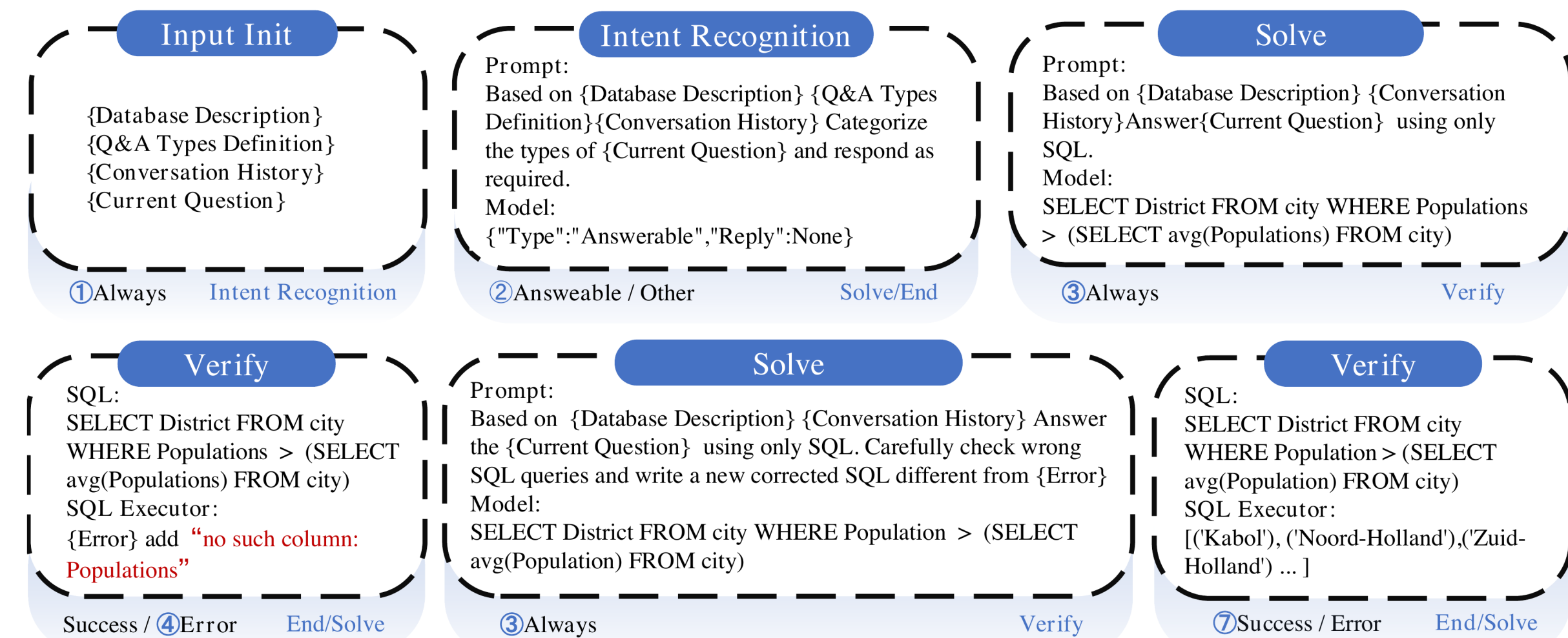


Fig. 3. Overview of StateFlow processes

Intent Recognition routes answerable questions to SQL generation and flags ambiguous, unanswerable, or improper requests. Verify checks execution and triggers correction when needed.

Fine-tuning improves SQL matching



Model	Sparc		CoSQL	
	QM	IM	QM	IM
GPT-3.5 Turbo	34.1	17.1	32.2	10.4
GPT-4 Turbo	31.4	18.8	33.6	10.8
Gemini-1.5 pro	36.6	18.3	36.7	9.9
Nsql-7b	5.6	0.2	3.4	0.0
+ QDA-SQL	51.4	32.7	46.9	19.8
Sqlcoder-7b-2	0.5	0.0	0.7	0.0
+ QDA-SQL	53.9	35.8	48.1	22.2
Codellama-7b	20.6	2.1	6.4	1.4
+ QDA-SQL	56.2	38.2	50.9	24.2
Codegemma-7b	29.7	12.6	24.5	6.8
+ QDA-SQL	57.6	42.4	56.5	28.7

GAINS ACROSS FOUR OPEN MODELS

CoSQL Question Match improves for every 7B model after dialogue augmentation.

CoSQL QM gain
NSQL-7B +43.5
SQLCoder-7B-2 +47.4
CodeLlama-7B +44.5
CodeGemma-7B +32.0

Sparc QM also rises for all four. SQLCoder gains 53.4 points; tuned CodeGemma leads CoSQL QM at 56.5.

+47.4 SQLCoder-7B-2 CoSQL Question Match

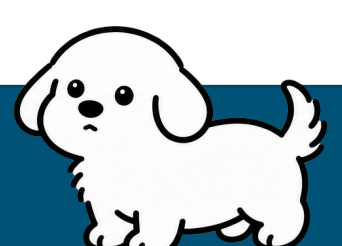
QDA-SQL lifts the score from 0.7 to 48.1, showing that dialogue augmentation can transform a compact model from near-zero matching to strong question-level retrieval.

+21.9 CodeGemma-7B CoSQL Interaction Match

The score rises from 6.8 to 28.7. This shows the gains extend from isolated question matching to full multi-turn interactions.

The gains are consistent across model families: the same augmentation helps both turn-level question matching and full-interaction matching, including for compact 7B backbones.

Intent recognition across four question types



Model	Precision	Recall	F1
SFT			
Answerable	82.4	99.3	90.1
Unanswerable	0	0	0
Ambiguous	6.5	0.9	1.6
Improper	100.0	79.5	88.6
average	47.2	44.9	45.1
QDA-SQL (With Intent)			
Answerable	90.5	94.8	92.6
Unanswerable	21.1	19.0	20.0
Ambiguous	30.8	14.8	20.0
Improper	97.5	99.7	98.6
average	60.0	57.1	57.8
QDA-SQL (SQL-only)			
Answerable	88.5	98.0	93.0
Unanswerable	0	0	0
Ambiguous	3.2	0.9	2.4
Improper	100.0	85.1	92.0
average	47.9	46.2	46.8

INTENT SIGNALS MATTER

With intent labels, average F1 reaches 57.8, versus 45.1 for SFT.

Answerable 90.1 to 92.6
Unanswerable 0 to 20.0
Ambiguous 1.6 to 20.0
Improper 88.6 to 98.6

SQL-only training misses ambiguous and unanswerable requests. Intent supervision teaches when to clarify or decline.

GitHub Repo: QDA-SQL

We propose a Text-to-SQL data augmentation tool.



Efficiently creating diverse multi-turn Text-to-SQL training samples in

3 steps!

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